Event-Level Decisions on a Self-Trained Audio-Transformer Ensemble for Noise-Event Detection in Vaani Field Speech

Abhilash Jacob Mathew
Team Arcturus (leaderboard: asymptoticchase), team leader
HCLTech
abhilashjacob16@gmail.com

Aman Kumar
Team Arcturus (leaderboard: asymptoticchase)
KIET GAZIABAD
aman.kumar.cse@kiet.edu

***Abstract*. Track 1 of the IndoML 2026 Datathon asks for the onset and offset of every noise event in recordings from the Vaani corpus, scored as Event F1 plus Segment Dice. We found that the two metrics reward different things: Event F1 counts events, whereas Dice counts frames, and most natural-recording events are short, so they carry many events but few frames. Our system therefore has two parts. A ten-member frame ensemble built on ATST-Frame, Dasheng and BEATs proposes candidate events. It is trained with Mean-Teacher and interpolation consistency on 80k unlabelled Vaani clips, an onset/offset-weighted loss, and one round of self-training in which the teacher is the full system. A second, event-level stage then decides which proposals to keep and which to merge, using boundary refiners as voters, gradient-boosted verifiers and a DETR-style set detector. A single confidence rule lets the same pipeline handle natural and synthetic clips. The final system scored 1.5893 (Event F1 0.7691, Dice 0.8202) and placed eighth on the Phase 2 leaderboard (as asymptoticchase).**

***Index Terms*. sound event detection, semi-supervised learning, self-training, ensemble, Vaani**

I. Training Data and Preprocessing

Task and metric. Each clip must be labelled with a list of noise events (breathing, horns, animals, alarms, music, machines, crying babies). A predicted event counts for Event F1 only when both its onset and its offset fall within $\max(0.2d, 0.05\text{s})$ of a reference of duration d (one-to-one matching, micro-averaged over all clips). Segment Dice is computed per clip on 10 ms frames and then averaged over clips. References are the union of all annotated categories. The Phase 2 test set has 3,437 natural recordings and 2,080 synthetic mixtures.

Vaani data. We used the Vaani Noise-Event dataset [1] from Hugging Face in all three of its tiers. Gold (`verified_timestamps`) has 11,111 clips with 19,957 verified events; the median event lasts 0.45 s, and breathing alone accounts for about 45% of the tags. Silver (`unverified_timestamps`, 61,642 clips) has freelancer timestamps that we measured to be considerably noisier than Gold, so we used Silver only as unlabelled audio and as speech backgrounds. Bronze (`no_timestamps`, 17,884 clips) has clip tags only and was also used as unlabelled audio. A fixed 20% of Gold (2,223 clips) was held out for fitting the post-processing models. The official validation set (1,806 natural and 695 synthetic clips) was first used for measurement on a seeded 70/30 split; after the organisers confirmed it was allowed, it was also used for fine-tuning. No test audio was used for training. No external datasets were used; the only outside resources are the public pretrained backbones in Table I.

Preprocessing. Audio is resampled to 16 kHz mono and padded or cut to 10 s, which gives 1,000 target frames at a 10 ms hop. Models that need a spectrogram use an 80-band log-mel ($n_{\text{fft}} = 1024$, hop 160). Clips longer than 10 s (13% of the test set) are processed in 10 s windows with 5 s overlap, and frame outputs are averaged where windows overlap.

Augmentation pipeline. (1) *Synthetic mixtures built from Vaani audio* [2]. We cut 10,546 noise events out of Gold clips using their verified timestamps, trimming near-silence at the ends. Breaths were left out because they cannot be separated from speech. Each synthetic clip takes a 10 s Silver background (peak-normalised to 0.7) and adds 1–3 events of 0.15–6 s at random non-overlapping positions at least 0.2 s apart. Each event is scaled to an SNR drawn uniformly from 0–15 dB relative to the background and faded in and out over 5 ms; its label is the exact mixing interval. We generated 15,000 such clips. (2) *Waveform mixup* for all members: two clips in a batch are blended with weight $\lambda \sim \text{Beta}(0.2, 0.2)$ with probability 0.5, and their label masks are combined by element-wise maximum. (3) *Spectral augmentation* for the two ATST members: FilterAugment [3] (2–5 bands, ±6 dB) and SpecAugment [4] (up to two frequency masks of $\leq$ 8 bins and two time masks of $\leq$ 20 frames). In the later fine-tuning phases these members also used synthetic device-impulse-response convolution and ±10% frequency warping, each with probability 0.5. (4) *Pseudo-labels* for Silver and Bronze, described in Section II-C.

II. Proposed Method

A. Frame Ensemble

Every member maps a 10 s waveform to 1,000 frame logits. A pretrained audio transformer produces tokens at 25 Hz. These are projected to 144 dimensions, interpolated to 100 Hz, and passed through a four-layer Conformer [5] (4 heads, depthwise kernel 31) and a linear frame classifier. The two ATST-Frame [6] members also have a parallel CNN branch on the log-mel. It pools only along frequency, so it keeps true 10 ms resolution, and its features are concatenated before the Conformer. The four Dasheng [7] and four BEATs [8]

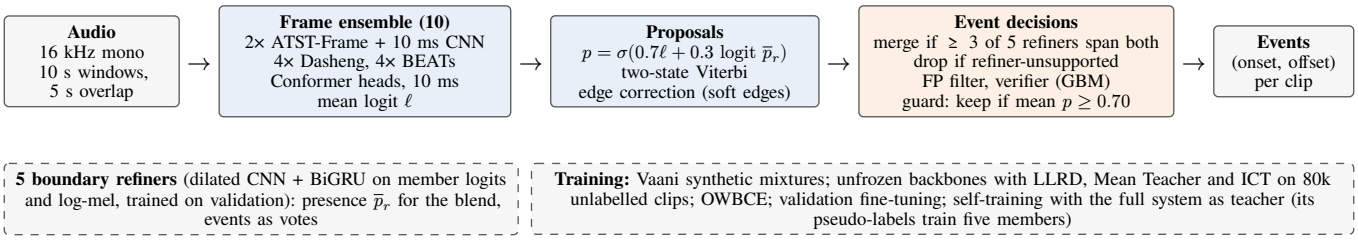


Fig. 1: Overview of the system. Every clip follows the same path; there is no natural/synthetic routing at inference.

members use the transformer alone. The ensemble output is the mean of the ten logits (Fig. 1). Five members come from the supervised and consistency phases below, and five are their self-trained students. Table I lists every pretrained model and checkpoint.

TABLE I: Pretrained models (backbones) and our checkpoints (listed with sha256 in the repository)

Name (version)	Source	Role
ATST-Frame (`ATST-F_strong_1`)	PretrainedSED [6]	2 members
BEATs (iter3+ AS2M)	microsoft/unilm [8]	4 members
Dasheng (`dasheng-base`)	HF mispeech [7]	4 members; detector
10 frame members	ours	frame ensemble
5 boundary refiners	ours, validation set	blend, votes
DETR-style detector	ours, Gold	verifier input
FP filter, verifier, PCAs	ours	event decisions
Edge-correction table	ours, Gold held-out	decoding

Losses. The frame loss is

$$\mathcal{L} = \mathcal{L}_{\text{OWBCE}} + 0.5\mathcal{L}_{\text{Dice}} + 0.2\,\text{MSE}(|\Delta p|, |\Delta y|), \quad (1)$$

where the last term matches the frame-to-frame changes of the predictions p to those of the targets y. We weighted the BCE term by onset and offset (OWBCE) [9]. Frame t has weight $w_t = 1 + hs(t)$, where s is a sine window of half-width 10 frames centred on every reference onset and offset, and $h = 4$; two Dasheng members were first trained with $h = 12$ and a 20-frame half-width. The weights are divided by their mean so that the loss scale does not change. On unlabelled audio we add the Mean-Teacher loss [10], the MSE between student probabilities and those of an EMA teacher with decay 0.999, and the Interpolation Consistency Training loss [11], which asks the student's output on a mixup of two clips ($\alpha = 0.2$) to match the same mixup of the teacher's outputs. Both are ramped in over five epochs with a sigmoid schedule.

B. Training Strategy

Training follows the stagewise recipe of the CPJKU DCASE 2024 system [12]. First, the backbone is frozen and the head is trained (AdamW, one-cycle schedule, peak learning rate 10^{-3}). Second, the backbone is unfrozen with layer-wise learning-rate decay over 12 groups ($10^{-4} \cdot 0.5^k$ from the top block), with the consistency losses switched on and early stopping on held-out data. This phase is what made members useful: frozen-feature members of the same backbones never improved the ensemble, while each unfrozen, consistency-regularised member added about 0.01 on the test set. Third, each member is fine-tuned for three epochs on Gold, synthetic data and the official validation set (backbone $2 \cdot 10^{-5}$, head 10^{-4}, batch shares 0.49 Gold, 0.15 synthetic, 0.21 validation-natural, 0.15 validation-synthetic). We keep the EMA teacher, which was usually the stronger model.

C. Self-Training With an Event-Level Teacher

Pseudo-labelling usually passes on the opinion of the frame model. We instead let the complete system of Sections II-A to II-D, including all event-level decisions, label the 79,526 Silver and Bronze clips (positive-frame rate 0.236). The five fine-tuned members were then trained for three more epochs with these labels making up 30% of the natural part of each batch. The resulting students had absorbed part of the event-level logic: 5% fewer clips needed event edits afterwards, and the frame model alone gained 0.0026 on the test set. Only inside this teacher did we restrict the event decisions to clips whose events have soft edges. The submitted system applies the same steps to every clip.

D. Proposals and Event-Level Decisions

Proposals. The ensemble logit ℓ is blended with the mean presence $\overline{p}_r$ of five boundary refiners: $p = \sigma(0.7\ell + 0.3 \text{ logit } \overline{p}_r)$. A two-state Viterbi decoder (stay probabilities 0.9802 and 0.9489) turns p into events. Natural references mark where a person heard a sound stop, and are tighter than the exact mixing times of the synthetic labels, so the model predicts short natural events too long. Each event's edge contrast, the probability jump across its first and last three frames, separates the two cases well (median 0.2 for natural and 0.8 for synthetic events). Events with contrast below 0.5 and duration under 1 s therefore have their edges moved by the median error of their duration bin, measured on held-out Gold.

Boundary refiners as voters. Each refiner reads the members' logits and the log-mel. It applies four dilated 1-D convolutions (width 128, dilations 1–8) and a two-layer BiGRU, and predicts presence, onset and offset impulses and the distance to each edge, in the style of RED/EPN [13] (BCE, focal and IoU losses). Five refiners were trained only on the validation set, whose annotation style is the closest to the test. Used directly as a decoder, a refiner gave up more Dice than it gained in F1. Used as voters on the ensemble's proposals they helped: two neighbouring events are merged when at least three of the five refiners have one event covering both, and an event that the first refiner does not overlap is dropped.

Verification. Two gradient-boosted classifiers estimate whether a proposal overlaps any reference. Both are trained on held-out Gold and validation clips, which the five base

members never saw, so their features mean the same on test. The false-positive filter uses 15 event statistics: duration, probability statistics, member agreement, edge contrast, gaps to neighbours and energy contrast. The verifier adds a 24-dimensional PCA of the members' pooled embeddings and the output of a DETR-style set detector [14]. The detector uses frozen Dasheng tokens, a mel stem, 10 anchor queries and Hungarian matching with an L1 term scaled by each reference's tolerance. The verifier's out-of-fold AUC is 0.868. Events above 0.2 (filter) or 0.25 (verifier) are dropped, and neighbours with a merge score above 0.6 are merged.

Confidence guard. Applied to every clip, these decisions at first cost 0.005, all of it on synthetic clips, where confident events were being dropped. Rather than route clips by domain, we added one rule for every clip: an event whose mean probability is at least 0.70 is never dropped. About 97% of events are protected this way, so drops act only on uncertain events.

Implementation. Python 3.12, PyTorch 2.13 (CUDA 13.0) and scikit-learn 1.9, on one NVIDIA RTX PRO 6000 (96 GB) and one A100 (80 GB). A member takes 2–6 GPU-hours to fine-tune. Inference on the full test set takes about 65 minutes on one GPU and needs about 9 GB of memory.

III. Results and Conclusion

Table II shows how the test score developed; each row was a separate submission. The final system scored **1.5893** (Event F1 0.7691, Dice 0.8202), **eighth on the Phase 2 leaderboard**, where our entry appears under the name asymptoticchase. On natural recordings it reached F1 0.4542 and Dice 0.7178, and on synthetic mixtures F1 0.9953 and Dice 0.990. Rerun from raw audio, 5,484 of the 5,517 test clips came out identical to the submitted predictions, and 28 more were within 10 ms.

TABLE II: Test-set results; each row adds one component ("–": not recorded)

System (Phase 2 test)	Comb.	F1	Dice	Nat. F1
From-scratch CNN-Conformer	1.0982	0.427	0.671	–
+ Vaani synthetic mixtures	1.4151	0.665	0.751	0.383
+ unfrozen backbones, MT + ICT	1.5427	0.733	0.810	0.412
+ OWBCE members	1.5545	0.736	0.819	0.420
+ edge corr., validation fine-tune	1.5694	0.747	0.822	0.425
+ refiner votes (merge, drop)	1.5793	0.759	0.820	0.437
+ FP filter and verifier	1.5819	0.761	0.821	0.438
+ event stage on long clips	1.5845	0.764	0.820	0.440
+ refiner blend	1.5865	–	–	–
+ self-trained students	1.5882	0.769	0.819	0.450
One pipeline, guard 0.70	1.5884	–	–	0.453
Final thresholds (submitted)	**1.5893**	**0.769**	**0.820**	**0.454**

The frame-level ingredients solved most of the synthetic half: synthetic F1 went from 0.44 to 0.995. The event-level stage brought the largest improvement on natural recordings (F1 0.425 to 0.454) without lowering their Dice. Refiner merges raised F1 and Dice together, as the metric analysis predicted. The guard sweep (0.90: 1.5858; 0.80: 1.5878; 0.70: 1.5883; 0.65: 1.5868) shows that one confidence rule can replace domain routing.

Several findings bound what is left to gain. Only 40% of natural validation references are matched within tolerance even though 92.6% are detected; the misses are mostly edges, with offsets running late. Reference edges sit at no consistent energy or probability level, which fits the large inter-annotator spread reported in the literature [15]. A tolerance-aware search for systematic edge shifts lost 0.002–0.005 out-of-fold. Larger or speech-oriented backbones (Dasheng-0.6B, Whisper, w2v-BERT, WavLM) and masked-reconstruction pretraining improved single models but moved the ensemble by only −0.0035 to +0.0018, and change-point decoding (cSEBB [16]) was worse. Dropping every uncertain event on test raised F1 by 0.0056 but cost 0.028 Dice because it emptied clips that contain noise, which shows the decision stage has reached its precision limit.

In conclusion, decisions about *which* events exist mattered more for this metric than refining *where* their edges lie. Self-training from an event-level teacher carried those decisions into the frame model, and a confidence guard let one pipeline serve both domains. The largest remaining gap is recall on speakers absent from training (63% of natural test clips against 37% in validation). Speaker-robust training and more diverse in-domain audio are the most promising next steps. Code and training scripts are at github.com/Abhilash1602/IndoML2026-Track1-Arcturus, and the checkpoints at huggingface.co/asymptoticlearning/IndoML2026-Track1-Arcturus.

References

[1] ARTPARK and IISc, "Vaani Noise Event Dataset." 2026.

[2] J. Salamon, D. MacConnell, M. Cartwright, P. Li, and J. P. Bello, "Scaper: A library for soundscape synthesis and augmentation," in *Proc. WASPAA*, 2017.

[3] H. Nam, S.-H. Kim, and Y.-H. Park, "FilterAugment: An Acoustic Environmental Data Augmentation Method," in *Proc. ICASSP*, 2022.

[4] D. S. Park *et al.*, "SpecAugment: A Simple Data Augmentation Method for Automatic Speech Recognition," in *Proc. Interspeech*, 2019.

[5] A. Gulati *et al.*, "Conformer: Convolution-augmented Transformer for Speech Recognition," in *Proc. Interspeech*, 2020.

[6] F. Schmid, T. Morocutti, F. Foscarin, J. Schlüter, P. Primus, and G. Widmer, "Effective Pre-Training of Audio Transformers for Sound Event Detection," *arXiv:2409.09546 (checkpoints: https://github.com/fschmid56/PretrainedSED)*, 2024.

[7] H. Dinkel, Z. Yan, Y. Wang, J. Zhang, Y. Wang, and B. Wang, "Scaling up masked audio encoder learning for general audio classification," in *Proc. Interspeech*, 2024.

[8] S. Chen *et al.*, "BEATs: Audio Pre-Training with Acoustic Tokenizers," in *Proc. ICML*, 2023.

[9] T. Song, "Onset and offset weighted loss function for sound event detection," *arXiv:2403.13254*, 2024.

[10] A. Tarvainen and H. Valpola, "Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results," in *Proc. NeurIPS*, 2017.

[11] V. Verma, A. Lamb, J. Kannala, Y. Bengio, and D. Lopez-Paz, "Interpolation Consistency Training for Semi-Supervised Learning," in *Proc. IJCAI*, 2019.

[12] F. Schmid, P. Primus, T. Morocutti, J. Greif, and G. Widmer, "Multi-Iteration Multi-Stage Fine-Tuning of Transformers for Sound Event Detection with Heterogeneous Datasets," *arXiv:2407.12997*, 2024.

[13] F. Schmid *et al.*, "Sound Event Detection with Boundary-Aware Optimization and Inference," *arXiv:2601.04178*, 2026.

[14] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-End Object Detection with Transformers," in *Proc. ECCV*, 2020.

[15] N. Koga, Y. Bando, and K. Imoto, "LEAD Dataset: How Can Labels for Sound Event Detection Vary Depending on Annotators?," *arXiv:2410.09778*, 2024.

[16] J. Ebbers, F. G. Germain, G. Wichern, and J. Le Roux, "Sound Event Bounding Boxes," in *Proc. Interspeech*, 2024.